P2 Chapter 11: Integration

Using Linear Substitutions

SKILL #2: Integrating $f(ax+b)$

$$\frac{d}{dx}(\sin(3x+1)) = 3\cos(3x+1)$$

Therefore:

$$\int \cos(3x+1)\ dx = \mathbf{\frac{1}{3}\sin(3x+1)+C}$$

✏ For any expression where inner function is $ax+b$, integrate as before and $\div\ a$.

$$\int f'(ax+b)dx = \frac{1}{a}f(ax+b)+c$$

Quickfire:

$$\int e^{3x}\ dx = \frac{1}{3}e^{3x}+C$$

$$\int \frac{1}{5x+2}dx = \frac{1}{5}\ln|5x+2|+C$$

$$\int 2\sec^2(3x-2)\ dx = \frac{2}{3}\tan(3x-2)+C$$

Tip: For $\int (ax+b)^n\ dx$, ensure you divide by the $(n+1)$ <u>and</u> the a

$$\int (3x+4)^3\ dx = \frac{1}{12}(3x+4)^4+C$$

$$\int \sin(1-5x)\ dx = \frac{1}{5}\cos(1-5x)+C$$

$$\int \frac{1}{3(4x-2)^2}dx = -\frac{1}{12}(4x-2)^{-1}+C$$

$$\int (10x+11)^{12} = \frac{1}{130}(10x+11)^{13}+C$$

Check Your Understanding

$$\int e^{3x+1}dx = \boxed{?}$$

$$\int \frac{1}{1-2x}dx = \boxed{?}$$

$$\int (4-3x)^5 dx = \boxed{?}$$

$$\int \sec(3x)\tan(3x)\,dx = \boxed{?}$$

Check Your Understanding

$$\int e^{3x+1}dx = \frac{\mathbf{1}}{\mathbf{3}}\boldsymbol{e}^{\mathbf{3}\boldsymbol{x}+\mathbf{1}} + \boldsymbol{C}$$

$$\int \frac{1}{1-2x}dx = -\frac{\mathbf{1}}{\mathbf{2}}\mathbf{ln}|\mathbf{1}-\mathbf{2}\boldsymbol{x}| + \boldsymbol{C}$$

$$\int (4-3x)^5 dx = -\frac{\mathbf{1}}{\mathbf{18}}(\mathbf{4}-\mathbf{3}\boldsymbol{x})^{\mathbf{6}} + \boldsymbol{C}$$

$$\int \sec(3x)\tan(3x)\,dx = \frac{\mathbf{1}}{\mathbf{3}}\mathbf{sec}(\mathbf{3}\boldsymbol{x}) + \boldsymbol{C}$$

Exercise 11B

1 a $\int \sin(2x+1)\,dx =$?

c $\int 4e^{x+5}\,dx =$?

e $\int cosec^2 3x\,dx =$?

f $\int \sec 4x \tan 4x\ dx =$?

g $\int 3\sin\left(\frac{1}{2}x+1\right)dx =$?

h $\int cosec\ 2x \cot 2x\ dx =$?

2 a $\int e^{2x} - \frac{1}{2}\sin(2x-1)\,dx$

$=$?

b $\int (e^x+1)^2\,dx =$?

c $\int \sec^2 2x\,(1+\sin 2x)\,dx =$?

d $\int \frac{3-2\cos\left(\frac{1}{2}x\right)}{\sin^2\left(\frac{1}{2}x\right)}dx =$?

e $\int e^{3-x} + \sin(3-x) + \cos(3-x)\,dx$

$=$?

3 a $\int \frac{1}{2x+1}dx =$?

b $\int \frac{1}{(2x+1)^2}dx =$?

c $\int (2x+1)^2\,dx =$?

d $\int \frac{3}{4x-1}dx =$?

f $\int \frac{3}{(1-4x)^2}dx =$?

h $\int \frac{3}{(1-2x)^3}dx =$?

j $\int \frac{5}{3-2x}dx =$?

4 a $\int 3\sin(2x+1) + \frac{4}{2x+1}\,dx$

$=$?

c $\int \frac{1}{\sin^2 2x} + \frac{1}{1+2x} + \frac{1}{(1+2x)^2}dx$

$=$?

Exercise 11B

1 a $\int \sin(2x+1)\,dx = -\frac{1}{2}\boldsymbol{cos(2x+1)} + \boldsymbol{C}$

c $\int 4e^{x+5}\,dx = \boldsymbol{4e^{x+5}} + \boldsymbol{C}$

e $\int cosec^2 3x\,dx = -\frac{1}{3}\boldsymbol{\cot 3x} + \boldsymbol{C}$

f $\int \sec 4x \tan 4x\ dx = \frac{1}{4}\boldsymbol{\sec 4x} + \boldsymbol{C}$

g $\int 3\sin\left(\frac{1}{2}x+1\right)dx = \boldsymbol{-6\cos\left(\frac{1}{2}x+1\right)}$

h $\int cosec\ 2x \cot 2x\ dx = -\frac{1}{2}\boldsymbol{cosec\ 2x} + \boldsymbol{C}$

2 a $\int e^{2x} - \frac{1}{2}\sin(2x-1)\,dx$

$= \frac{1}{2}\boldsymbol{e^{2x}} + \frac{1}{4}\boldsymbol{\cos(2x-1)} + \boldsymbol{C}$

b $\int (e^x+1)^2\,dx = \frac{1}{2}\boldsymbol{e^{2x}} + \boldsymbol{2e^x} + \boldsymbol{x} + \boldsymbol{C}$

c $\int \sec^2 2x\,(1+\sin 2x)\,dx = \frac{1}{2}\boldsymbol{\tan 2x} + \frac{1}{2}\boldsymbol{\sec 2x}$

d $\int \frac{3-2\cos\left(\frac{1}{2}x\right)}{\sin^2\left(\frac{1}{2}x\right)}dx = \boldsymbol{-6\cot\left(\frac{1}{2}x\right)} + \boldsymbol{4cosec\left(\frac{1}{2}x\right)}$

e $\int e^{3-x} + \sin(3-x) + \cos(3-x)\,dx$

$= \boldsymbol{-e^{3-x} + \cos(3-x) - \sin(3-x)} + \boldsymbol{C}$

3 a $\int \frac{1}{2x+1}dx = \frac{1}{2}\boldsymbol{\ln|2x+1|} + \boldsymbol{C}$

b $\int \frac{1}{(2x+1)^2}dx = -\frac{1}{\boldsymbol{2(2x+1)}} + \boldsymbol{C}$

c $\int (2x+1)^2\,dx = \frac{1}{6}\boldsymbol{(2x+1)^3} + \boldsymbol{C}$

d $\int \frac{3}{4x-1}dx = \frac{3}{4}\boldsymbol{\ln|4x-1|} + \boldsymbol{C}$

f $\int \frac{3}{(1-4x)^2}dx = \frac{\boldsymbol{3}}{\boldsymbol{4(1-4x)}} + \boldsymbol{C}$

h $\int \frac{3}{(1-2x)^3}dx = \frac{\boldsymbol{3}}{\boldsymbol{4(1-2x)^2}} + \boldsymbol{C}$

j $\int \frac{5}{3-2x}dx = -\frac{5}{2}\boldsymbol{\ln|3-2x|} + \boldsymbol{C}$

4 a $\int 3\sin(2x+1) + \frac{4}{2x+1}\,dx$

$= -\frac{3}{2}\boldsymbol{\cos(2x+1)} + \boldsymbol{2\ln|2x+1|} + \boldsymbol{C}$

c $\int \frac{1}{\sin^2 2x} + \frac{1}{1+2x} + \frac{1}{(1+2x)^2}dx$

$= -\frac{1}{2}\boldsymbol{\cot 2x} + \frac{1}{2}\boldsymbol{\ln|1+2x|} - \frac{\boldsymbol{1}}{\boldsymbol{2(1+2x)}}$

Exercise 11.2

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Homework Exercise

1 Integrate the following:

a $\sin(2x+1)$ **b** $3e^{2x}$

c $4e^{x+5}$ **d** $\cos(1-2x)$

e $\operatorname{cosec}^2 3x$ **f** $\sec 4x \tan 4x$

g $3\sin\left(\frac{1}{2}x+1\right)$ **h** $\sec^2(2-x)$ **i** $\operatorname{cosec} 2x \cot 2x$ **j** $\cos 3x - \sin 3x$

Hint For part **a** consider $y = \cos(2x+1)$. You do not need to write out this step once you are confident with using this method.

2 Find the following integrals.

a $\int \left(e^{2x} - \frac{1}{2}\sin(2x-1)\right) dx$ **b** $\int (e^x+1)^2\, dx$

c $\int \sec^2 2x(1+\sin 2x)\, dx$ **d** $\int \frac{3-2\cos\frac{1}{2}x}{\sin^2\frac{1}{2}x}\, dx$

e $\int \left(e^{3-x} + \sin(3-x) + \cos(3-x)\right) dx$

3 Integrate the following:

a $\frac{1}{2x+1}$ **b** $\frac{1}{(2x+1)^2}$ **c** $(2x+1)^2$ **d** $\frac{3}{4x-1}$

e $\frac{3}{1-4x}$ **f** $\frac{3}{(1-4x)^2}$ **g** $(3x+2)^5$ **h** $\frac{3}{(1-2x)^3}$

4 Find the following integrals.

a $\int \left(3\sin(2x+1) + \frac{4}{2x+1}\right) dx$ **b** $\int \left(e^{5x} + (1-x)^5\right) dx$

c $\int \left(\frac{1}{\sin^2 2x} + \frac{1}{1+2x} + \frac{1}{(1+2x)^2}\right) dx$ **d** $\int \left((3x+2)^2 + \frac{1}{(3x+2)^2}\right) dx$

Homework Exercise

5 Evaluate:

a $\int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} \cos(\pi - 2x)\,dx$ **b** $\int_{\frac{1}{2}}^{1} \frac{12}{(3-2x)^4}\,dx$ **c** $\int_{\frac{2\pi}{9}}^{\frac{5\pi}{18}} \sec^2(\pi - 3x)\,dx$ **d** $\int_{2}^{3} \frac{5}{7-2x}\,dx$

6 Given $\int_{3}^{b} (2x-6)^2\,dx = 36$, find the value of b. **(4 marks)**

7 Given $\int_{e^2}^{e^8} \frac{1}{kx}\,dx = \frac{1}{4}$, find the value of k. **(4 marks)**

8 Given $\int_{\frac{\pi}{4k}}^{\frac{\pi}{3k}} (1 - \pi \sin kx)\,dx = \pi(7 - 6\sqrt{2})$, find the exact value of k. **(7 marks)**

Problem-solving

Calculate the value of the indefinite integral in terms of k and solve the resulting equation.

Challenge

Given $\int_{5}^{11} \frac{1}{ax+b}\,dx = \frac{1}{a}\ln\left(\frac{41}{17}\right)$, and that a and b are integers with $0 < a < 10$, find two different pairs of values for a and b.

Homework Answers

1 **a** $-\frac{1}{2}\cos(2x+1)+c$ **b** $\frac{3}{2}e^{2x}+c$

c $4e^{x+5}+c$ **d** $-\frac{1}{2}\sin(1-2x)+c$

e $-\frac{1}{3}\cot 3x+c$ **f** $\frac{1}{4}\sec 4x+c$

g $-6\cos\left(\frac{1}{2}x+1\right)+c$ **h** $-\tan(2-x)+c$

i $-\frac{1}{2}\operatorname{cosec} 2x+c$ **j** $\frac{1}{3}(\sin 3x+\cos 3x)+c$

2 **a** $\frac{1}{2}e^{2x}+\frac{1}{4}\cos(2x-1)+c$ **b** $\frac{1}{2}e^{2x}+2e^{x}+x+c$

c $\frac{1}{2}\tan 2x+\frac{1}{2}\sec 2x+c$

d $-6\cot\left(\frac{1}{2}x\right)+4\operatorname{cosec}\left(\frac{1}{2}x\right)+c$

e $-e^{3-x}+\cos(3-x)-\sin(3-x)+c$

3 **a** $\frac{1}{2}\ln|2x+1|+c$ **b** $-\dfrac{1}{2(2x+1)}+c$

c $\dfrac{(2x+1)^3}{6}+c$ **d** $\frac{3}{4}\ln|4x-1|+c$

e $-\frac{3}{4}\ln|1-4x|+c$ **f** $\dfrac{3}{4(1-4x)}+c$

g $\dfrac{(3x+2)^6}{18}+c$ **h** $\dfrac{3}{4(1-2x)^2}+c$

4 **a** $-\frac{3}{2}\cos(2x+1)+2\ln|2x+1|+c$

b $\frac{1}{5}e^{5x}-\dfrac{(1-x)^6}{6}+c$

c $-\frac{1}{2}\cot 2x+\frac{1}{2}\ln|1+2x|-\dfrac{1}{2(1+2x)}+c$

d $\dfrac{(3x+2)^3}{9}-\dfrac{1}{3(3x+2)}+c$

5 **a** 1 **b** $\frac{7}{4}$ **c** $\dfrac{2\sqrt{3}}{9}$ **d** $\frac{5}{2}\ln 3$

6 $b=6$ **7** $k=24$ **8** $k=\frac{1}{12}$

Challenge

$a=4,\ b=-3$ or $a=8,\ b=-6$